

Cryptography & Cyber Law

Course Code: ICT 4115

1. Abstract Algebra & Number Theory fundamental:

(i) Fermat's Little Theorem (FLT):

If p is a prime number and $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof Sketch: The non-zero elements of $\mathbb{Z}_p$ form a multiplicative group of order $p-1$. By Lagrange's Theorem, the order of any element divides the order of the group.

$$\text{hence, } a^{p-1} \equiv 1 \pmod{p}$$

Miller-Rabin Primality Test:

When $n-1 = 2^s \cdot d$, where d is odd, for

a base $a \in [2, n-2]$:

(i) compute $x = a^d \pmod{n}$

(ii) If $x \equiv 1 \pmod{n}$ or $x \equiv -1 \pmod{n}$

(iii) Otherwise, square x up to $s-1$

times: $x \leftarrow x^2 \pmod{n}$. If $x \equiv -1 \pmod{n}$ at any step, n passes.

(iv) If $-1 \pmod{n}$ is never encountered, n is definitely composite.

Evaluating:

$$n = 561 \quad (561 = 3 * 11 * 17)$$

$$n-1 = 560 = 2^4 \cdot 35 \Rightarrow s = 4, d = 35$$

Testing with base $a = 2$:

- $2^{35} \equiv 263 \pmod{561}$ (neither 1 nor -1)
- $263^2 \equiv 166 \pmod{561}$
- $166^2 \equiv 67 \pmod{561}$
- $67^2 \equiv 1 \pmod{561}$

Since 1 was reached without immediately being preceded by -1, 561 is composite by Miller-Rabin.

However, evaluating simple FLT gives

$$2^{560} \equiv 1 \pmod{561}, \text{ because}$$

$$a^{560} \equiv 1 \pmod{561} \text{ holds for every } a$$

$$(a, 561) = 1$$

So, 561 is a Carmichael number

2. $\mathbb{Z}_n$ is a field if and only if n is prime

If n is prime

for any non-zero element $a \in \mathbb{Z}_p$, $\gcd(a, p) = 1$. By Bezout's identity, there exist integers x, y such that $ax + py = 1 \Rightarrow ax \equiv 1 \pmod{p}$.

Thus, every non-zero element has a multiplicative inverse, making $(\mathbb{Z}_p, +, \cdot)$ a field.

If n is composite:

n can be factored as $n = a \cdot b$ where $1 < a, b < n$. In $\mathbb{Z}_n$,

$$a \cdot b \equiv 0 \pmod{n} \text{ with } a \not\equiv 0 \pmod{n} \text{ and } b \not\equiv 0 \pmod{n}$$

Examples:

- $\mathbb{Z}_{11}$ is field, because 11 is prime
- $\mathbb{Z}_{12}$ is not a field because $3 \times 4 \equiv 0 \pmod{12}$, meaning neither 3 nor 4 possesses an inverse.

3. Construct of $\mathbb{GF}(2^3)$

- Irreducible polynomial: $f(x) = x^3 + x + 1$ over $\mathbb{GF}(2)$.
- Let α be a root of $f(x)$, so $\alpha^3 + \alpha + 1 = 0 \Rightarrow \alpha^3 = \alpha + 1$.

Field Elements:

The 8 elements are polynomial of degree < 3

$$GF(2^3) = \{0, 1, \alpha, \alpha^2, \alpha+1, \alpha^2+\alpha, \alpha^2+1, \alpha^2+\alpha+1\}$$

Cyclic Multiplicative Group:

The multiplicative group $GF(2^3) \setminus \{0\}$ has order $2^3 - 1 = 7$

- $\alpha^1 = \alpha$
- $\alpha^2 = \alpha^2$
- $\alpha^3 = \alpha^2 + 1$
- $\alpha^4 = \alpha^2 + \alpha$
- $\alpha^5 = \alpha^3 + \alpha^2 = \alpha^2 + \alpha + 1$
- $\alpha^6 = \alpha^3 + \alpha^2 + \alpha = \alpha^2 + 1$
- $\alpha^7 = \alpha^3 \cdot \alpha^4 = 1$

Because $\alpha^k \neq 1$ for $1 \leq k < 7$, α has order 7. Thus, $GF(2^3)^*$ is cyclic & generated by α :
 $\alpha^i, \alpha^j = \alpha^{i+j \bmod 7}$

4. Square and multiply complexity

For an exponent e , its binary representation contains $k = \lceil \log_2 e \rceil + 1$ bits

(i) Number of squaring: $\lceil \log_2 e \rceil$

(ii) Multiplication: $\lceil \log_2 e \rceil$

(iii) Total time complexity: $O(\log e)$

Evaluating $7^{560} \pmod{561}$:

Since 561 is a Carmichael number and $\gcd(7, 561) = 1$

$$7^{560} \equiv 1 \pmod{561}$$

Result = 1

Shift Rows:

The first row (Row) shifts left by 0 positions:

$$\text{Row } 0 = [6D, 63, 2E, CB]$$

18. AES-128 Key Schedule:

Given $K_0 =$

00	01	02	03	04	05	06
07	08	09	0A	0B	0C	0D
0E	0F					

$$w_0 = 00010203, \quad w_1 = 04050607, \\ w_2 = 08090A0B, \quad w_3 = 0C0D0E0F$$

To compute w_4 :

(i) Rotword (w_3) = 0D0E0F0C

(ii) Subword = D7AB76FE

(iii) $Rcon_1 = 01000000$

(iv) $w_4 = w_0 \oplus \text{Subword} \oplus Rcon_1$

Subsequent words:

4 $\Rightarrow$ AES: Modes, Transformations & Side channels.

16. AES Modes of Operation

Mode	ENCRYP. Parallel	Parallel DECRYP.	Bit Error Propagation	Nonce/IV Req.
ECB	Yes	Yes	Isolated to single block	None
CBC	No	Yes	current block 1 bit next block	Unique
CFB	No	Yes	current bit 1 whole next block	Unique IV
OFB	No	No	Exactly 1 bit	Never reuse nonce
CTR	Yes	Yes	Exactly 1 bit	strict

① live video streaming

② static Disk Image

17. AES SubBytes & ShiftRows:

Given the first state row: $[23, A7, A7, 19]$

• SubBytes:

$S(23) = 6D$, $S(A7) = 63$, $S(40) = 2E$

$S(19) = C6$

Resulting Row: $[6D, 63, 2E, C6]$

- It guarantees high diffusion and confusion, preventing differential cryptanalysis.

15. Password Brute-Force Analysis

Assume cracking rate: 10^9 guesses/sec

(i) 8 character alphanumeric password

(a-z, A-Z, 0-9) $\Rightarrow 62$ symbols

$$N = 62^8 = 218,340,105,584,896$$

Combinations

$$\text{Worst-case-time} = \frac{2.183 \times 10^{14}}{10^9} \approx 218,340.5$$

$$\approx 60.65 \text{ Hours}$$

$$\approx 2.53 \text{ days}$$

$$\text{Average-case-time} = \frac{\text{Worst-case}}{2} \approx 30.33 \text{ hours}$$

(ii) 6 Digit PIN [0-9] $\Rightarrow 10^6$ combinations

$$\text{Worst-case-time: } \frac{10^6}{10^9} = 0.001 \text{ sec.}$$

$$\text{Average-time: } 0.0005 \text{ sec.}$$

$$T_0 = 11111111 \oplus 11111111 = 555555$$

AP ← signal
 $\therefore F0 \oplus F0 = FF$ and $AA \oplus FF = 55$

Round 2:

$$L_2 = R_1 = 55555555$$

$$\begin{aligned} R_2 &= L_1 \oplus F(R_1, K_2) \\ &= F0F0F0F0 \oplus (55555555 \oplus 33333333) \\ &= 96969696 \end{aligned}$$

13. Avalanche Effect

A desirable property of cryptographic algorithms where a change in a single bit in either the plaintext or key causes approximately half of the ciphertext bits to change:

$$\frac{\Delta \text{ output bits}}{\text{Total output bits}} \approx 50\%$$

Example (EN): I is at (3,4) N is at (1,3) $\rightarrow$ Form rectangle $\rightarrow$ QA.

12. Playfair weaknesses

- Digraph Frequency Leakage

- Small Alphabet space

- Symmetry Artifacts

13. Two Feistel Rounds

Given: $F(R, K) = R \oplus K \oplus A$

$L_0 = AAAAAAAAAA_{16}$, $R_0 = FFFFFFFF_{16}$

$K_1 = 0F0F0F0F_{16}$, $K_2 = 33333333_{16}$

• Round 1: $R_0 = FFFFFFFF$

$L_1 =$

$R_1 = L_0 \oplus F(R_0, K_1)$

$= AAAAAAAAAA \oplus (FFFFFFF \oplus FFFFFFFF)$

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classical Symmetric Block Ciphers

11. Playfair cipher

Keyword : MONARCHY (combine I/J)

5x5 key Matrix :

M O N A R

C H Y B D

E F G I K

L P Q S T

U V W X Z

Plaintext : INFORMATION SECURITY

- Digraph preparation:

I N F O R M A T I O N S E C U R I T Y

- Encryption Rules:

(i) Rectangle $\rightarrow$ swap columns

(ii) Same row $\rightarrow$ right shift

(iii) Same column $\rightarrow$ shift down

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10. Modular Multiplicative Inverse:

(EEA vs Fermat)
compute $3^{-1} \pmod{7}$:

Method 1: Extended Euclidean Algorithm

$$7 = 2(3) + 1 \Rightarrow 1 = 1(7) - 2(3)$$

$$\Rightarrow -2 \times 3 \equiv 1 \pmod{7}$$

$$3^{-1} \equiv -2 \equiv 5 \pmod{7}$$

Method 2: Fermat's Little Theorem
($a^{p-1} \equiv 1 \pmod{p}$)
($a^{p-2} \equiv a^{-1} \pmod{p}$)

$$3^{7-2} = 3^5 = 243 \equiv 5 \pmod{7}$$

$$243 \equiv 34 \times 7 + 5 \Rightarrow 3^{-1} \equiv 5 \pmod{7}$$

Comparison:

EEA works for any modulus m

where $\gcd(a, m) = 1$ (FLT) strictly

requires m to be prime

9. Bellare fault attack on RSA-CRT

Suppose a hardware fault occurs during modular exponentiation modulo p , while the modulo q calculation completes correctly:

$$s_1' \not\equiv M^d \pmod{p}, \quad s_2 \equiv M^d \pmod{q}$$

The combined connected signature s' satisfies:

$$s' \equiv s \pmod{p} \Rightarrow s' - s \not\equiv 0 \pmod{p}$$

$$s' \equiv s \pmod{q} \Rightarrow s' - s \equiv 0 \pmod{q}$$

An attacker can factor the public modulus n instantly:

$$\gcd(s - s', n) = q \text{ or } \gcd((s - s')^e - M, n) = q$$

Countermeasures:

- (i) Signature verification
- (ii) Cryptographic blinding
- (iii) Double computation

8. CRT Based RSA Optimization:

In RSA decryption/signing ($M = c^d \pmod{n}$)

(i) Precompute:

$$d_P = d \pmod{p-1}, \quad d_Q = d \pmod{q-1}$$

$$q_{inv} = q^{-1} \pmod{p}$$

(ii) Compute two half-sized exponentiations

$$m_1 = c^{d_P} \pmod{p}, \quad m_2 = c^{d_Q} \pmod{q}$$

(iii) Recombine (using) Garner's formula:

$$h = q_{inv} (m_1 - m_2) \pmod{p}$$

$$\Rightarrow M = m_2 + h \cdot q$$

Speedup Analysis:

(i) standard exponentiation on k -bit numbers scales as $O(k^3)$ (or $O(k^2)$ using fast multiplication).

$$x \equiv 2 \pmod{3}, x \equiv 3 \pmod{5}, x \equiv 4 \pmod{7}$$

$$x \equiv 5 \pmod{11}$$

$$M = 3 \times 5 \times 7 \times 11 = 1155$$

$$M_1 = \frac{1155}{3} = 385 \Rightarrow 385 \equiv 1 \pmod{3} \Rightarrow y_1 \equiv 1 \pmod{3}$$

$$M_2 = \frac{1155}{5} = 231 \Rightarrow 231 \equiv 1 \pmod{5} \Rightarrow y_2 \equiv 1 \pmod{5}$$

$$M_3 = \frac{1155}{7} = 165 \Rightarrow 165 \equiv 4 \pmod{7}$$

$$\Rightarrow 4y_3 \equiv 1 \pmod{7} \Rightarrow y_3 \equiv 2 \pmod{7}$$

$$M_4 = \frac{1155}{11} = 105 \Rightarrow 105 \equiv 6 \pmod{11}$$

$$\Rightarrow 6y_4 \equiv 1 \pmod{11} \Rightarrow y_4 \equiv 2 \pmod{11}$$

compute x :

$$x = \sum a_i M_i y_i \pmod{M}$$

$$= 2(385)(1) + 3(231)(1) + 4(165)(2) + 5(105)(2) \pmod{1155}$$

$$= 3833 \pmod{1155}$$

$$= 368 \pmod{1155} \Rightarrow x \equiv 368 \pmod{1155}$$

Euclidean Division Theorem:

$$240 = 5 \times 46 + 10$$

$$46 = 4 \times 10 + 6$$

$$10 = 1 \times 6 + 4$$

$$6 = 1 \times 4 + 2$$

$$4 = 2 \times 2 + 0 \Rightarrow \gcd(240, 46) = 2$$

Back Substitution:

$$2 = 6 - 1(4)$$

$$= 6 - 1(10 - 1(6)) = 2(6) - 1(10)$$

$$= 6 - 1(10 - 1(6)) = 2(6) - 1(10)$$

$$= 2(46 - 4(10)) - 1(10) = 2(46) - 9(10)$$

$$= 2(46) - 9(240 - 5(46))$$

$$= -9(240) + 47(46)$$

$$x = -9, y = 47$$

Chinese Remainder Theorem:

7. Chinese Remainder Theorem:
(CRT)

Solve the system:

5. AKS VS. Miller-Rabin Primality Test :

Feature	AKS Algorithm	Miller-Rabin Algorithm
Type	Deterministic	Probabilistic
Error Probability	0	$\leq 4^{-K}$ (K independent bases)
Complexity	$O(\log^6 n)$	$O(K \cdot \log^3 n)$
Practical use	Rarely used	Standard
2048/4096-bit RSA	Infeasible in real-time	Extremely fast.

6. Modular Arithmetic & Chinese Remainder Theorem :

(i) Extended Euclidean Algorithm (EEA)

Find $\text{gcd}(240, 46)$ and integers x, y such that $240x + 46y = \text{gcd}(240, 46)$:

